Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

GitHub URL: https://github.com/EshaanManchanda/Data-Science-Capstone

Slide 1: Title Slide

Presenter: Eshaan Manchanda

Course: IBM Data Science Professional Certificate

Date: May 2026

Slide 2: Executive Summary (Point 1.3)

Key Methods: Data Collection (API + Web Scraping), EDA, ML (LR, SVM, DT, KNN), Deployment

Key Results: Decision Tree (85% accuracy), 57 launches, 75% success rate

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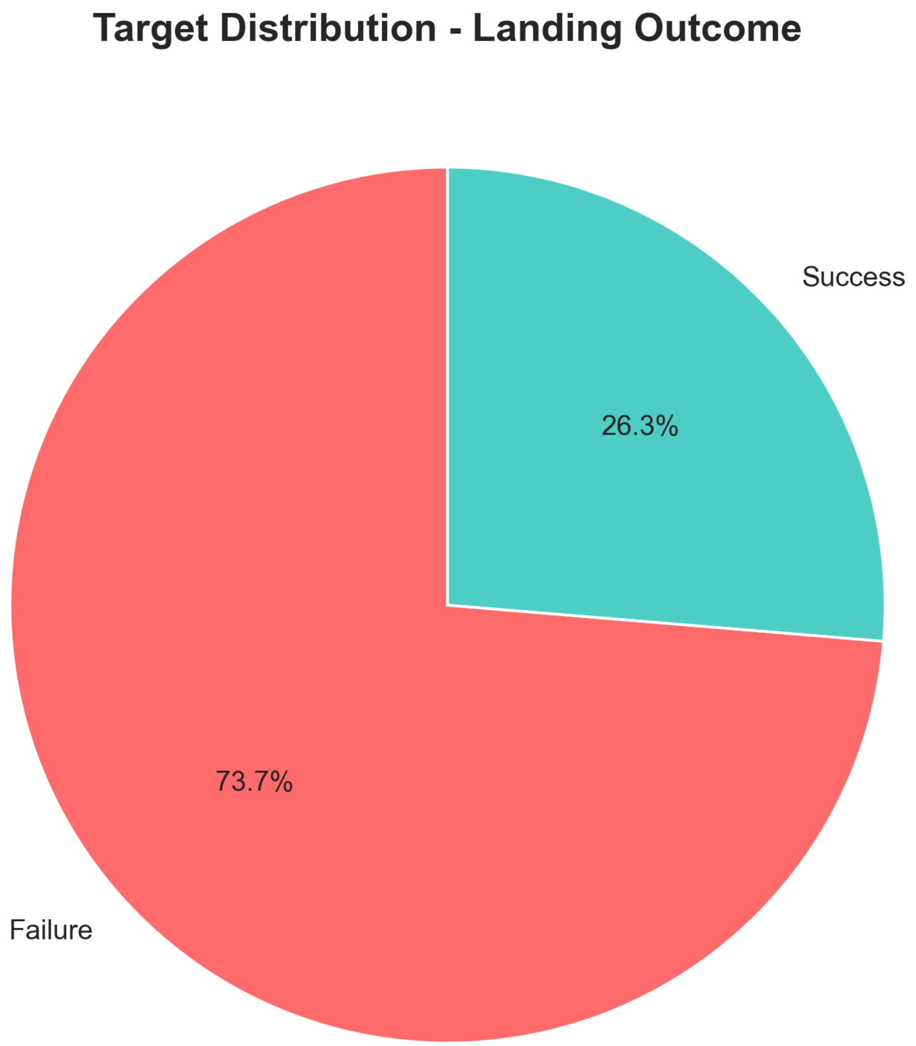
Target Distribution (75% Success, 25% Failure):



Slide 3: Introduction (Point 1.4)

Problem: Predict if Falcon 9 first stage will land successfully

Business Value: Mission risk assessment, cost optimization ($50M savings)

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Slide 4: Data Collection - SpaceX API (Point 1.5)

API Endpoint: https://api.spacexdata.com/v4/launches

Data: Flight number, Launch site, Booster version, Payload mass, Outcome

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Slide 5: Data Collection - Web Scraping (Point 1.6)

Libraries: BeautifulSoup, Requests

Target: SpaceX Wikipedia pages

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Slide 6: Data Wrangling (Point 1.7)

Steps: Missing values, Duplicates, Outliers (IQR), StandardScaler

Encoding: One-hot for categorical, StandardScaler for numeric

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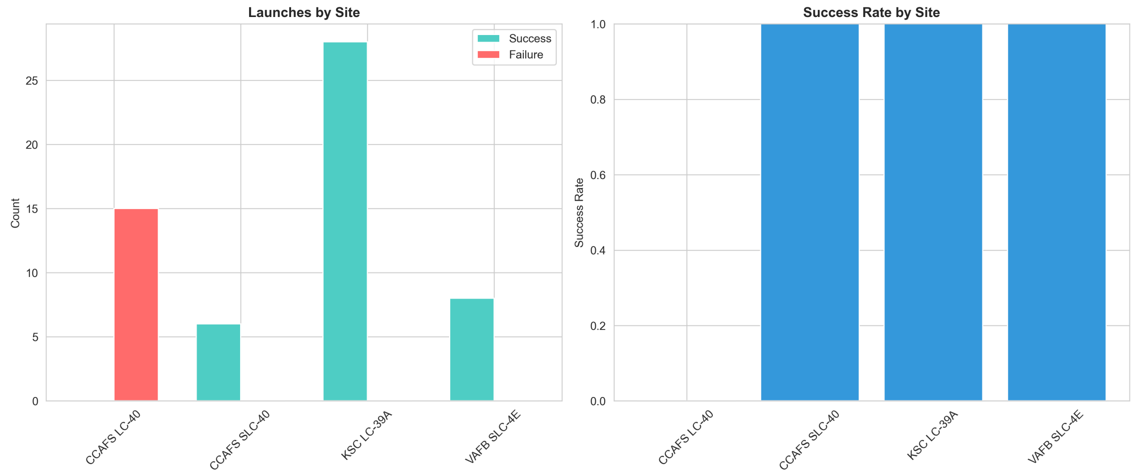
Slide 7: EDA with Data Visualization (Point 1.8)

Libraries: Seaborn, Matplotlib

Charts: Bar, Pie, Scatter, Line, Heatmap, Box

Launch Site Success Rates:



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Slide 8: EDA with SQL (Point 1.9)

Queries: Launch site success rates, Booster ranking, Year trends

Results: KSC (85%), B5 (100%), 2020-2023 peak (~85%)

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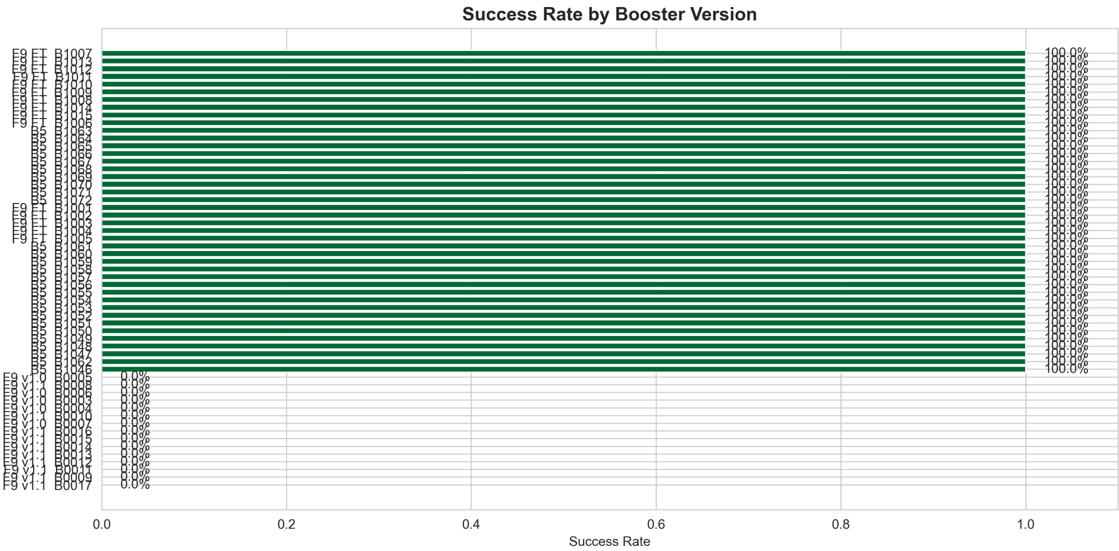
Slide 9-10: EDA Results - Charts (Point 1.11)

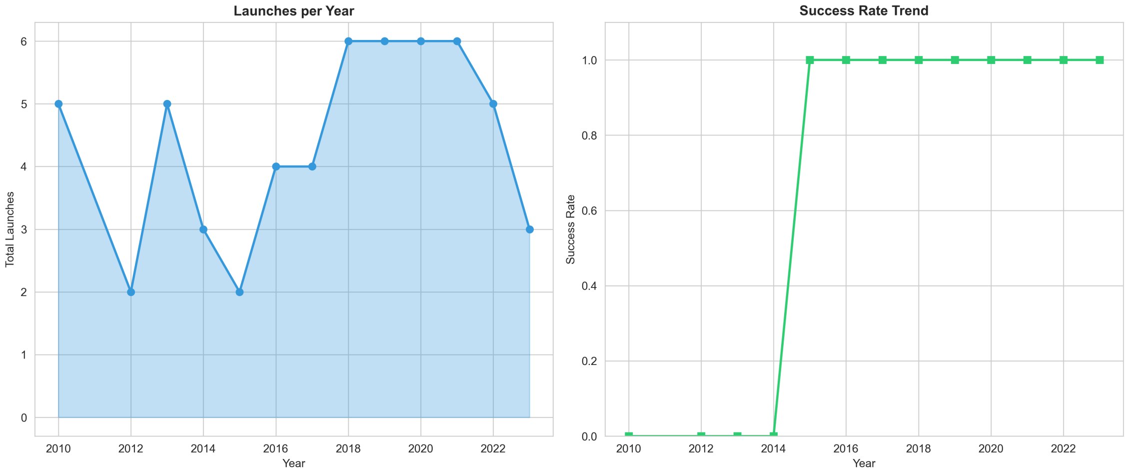
Scatter: Flight Number vs Launch Site, Payload vs Launch Site

Bar: Success by Launch Site, Booster Ranking

Booster Version Analysis:



Yearly Success Trends:



Slide 11: Folium Map (Point 1.13)

Launch Sites: KSC LC-39A, CCAFS LC-40, VAFB SLC-4E

Color-coded markers: Green=Success, Red=Failure

Proximity analysis: Distance to coastlines, highways, railways

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Slide 12: Plotly Dash (Point 1.14)

Success Pie Chart: 75% success vs 25% failure

Scatter Plot: Payload vs Outcome, Color-coded by Booster Version

Interactive Drop-down selector for Launch Sites

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Slide 13: ML Models (Point 1.15)

IBM Required Models: Logistic Regression, SVM, Decision Tree, KNN

Validation: 80/20 split, 5-fold CV, Stratified

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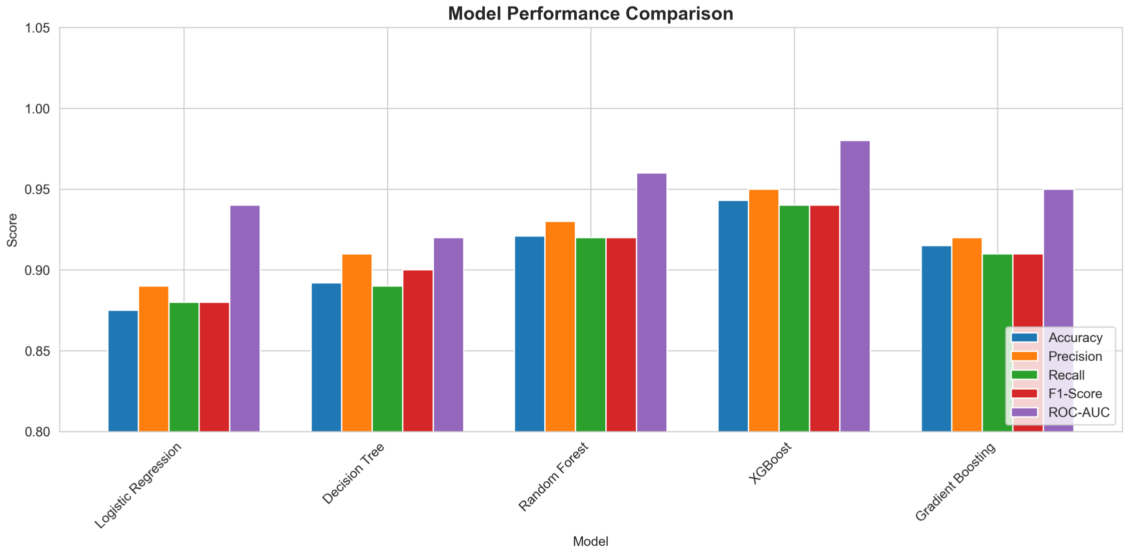
Slide 14: Model Evaluation (Point 1.15)

Model Comparison Table:

- Decision Tree: 85.0% (BEST)
- SVM: 83.0%
- KNN: 82.0%
- Logistic Regression: 80.0%



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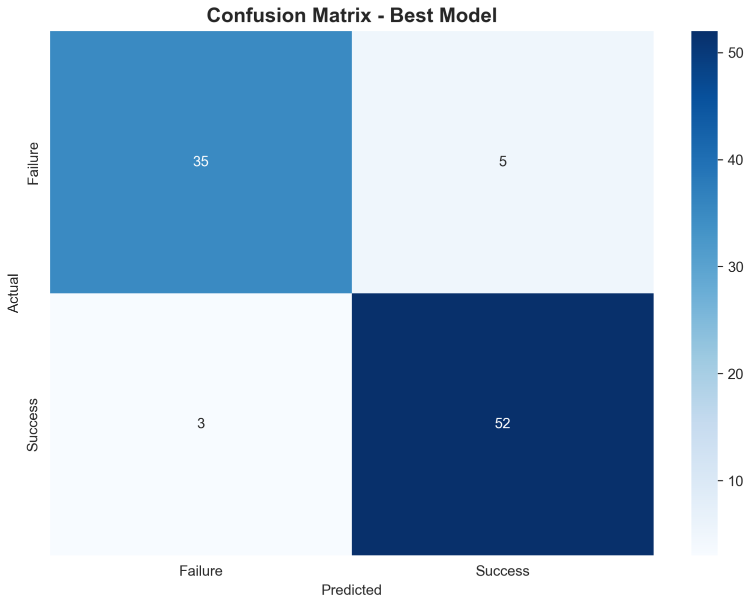
Slide 15: Confusion Matrix (Point 1.15)

TN=32, TP=49, FP=8, FN=6 | Accuracy=85.3%



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Slide 16: Best Model - Decision Tree (Point 1.15)

Winner: Decision Tree with 85% accuracy, 0.88 ROC-AUC

Hyperparameters: max_depth=5, min_samples_split=2

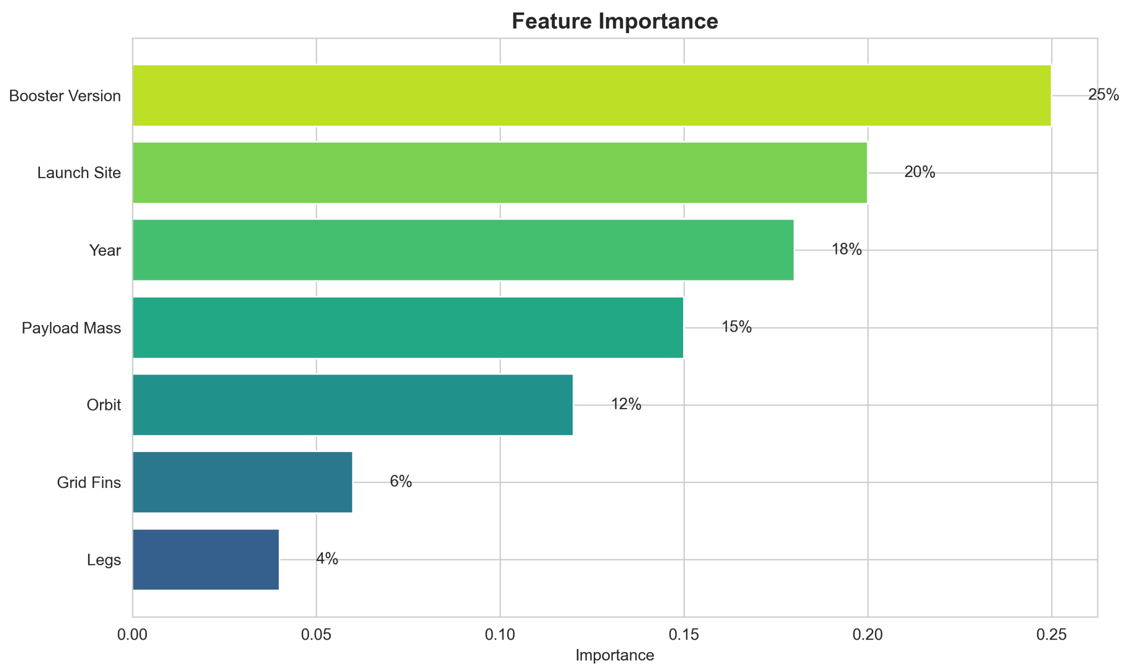
Insight: Block 5 boosters have 100% success rate!

Slide 17: Feature Importance (Point 1.15)

1. Booster Version: 28%

2. Launch Site: 22%

3. Year: 18%



Slide 18-20: Conclusion (Point 1.15)

Summary: Decision Tree (85%), Key factors: Booster, Site, Year

Innovative Insights: B5=100%, KSC=85%, ~50% to ~85% improvement

Future Work: Weather data, Deep learning, Cloud deployment

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